

BEMNET KIBRET

BACKEND & SYSTEMS ENGINEER

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PROFESSIONAL SUMMARY

Disciplined Backend & Systems Engineer specializing in relational database architecture, distributed microservices, and resilient cloud pipelines. Deep expertise in PostgreSQL (157 strict migrations, Row-Level Security, sub-100ms query plans), containerized Python/Node.js microservices on AWS (ECS, Lambda), and idempotent event-driven webhooks. Proven track record of designing fault-tolerant offline-first synchronization layers guaranteeing 0% data loss across distributed nodes. Passionate about system reliability, clean API boundaries, and automated CI/CD testing for high-concurrency environments.

EXPERIENCE

Lead Backend & Systems Engineer | Senselet (Enterprise SaaS Infrastructure) 2024 – Present

- **Mission-Critical Database Architecture:** Architected an offline-first distributed data layer utilizing PostgreSQL Dual-ID resolution; authored 157 strict migrations with organization-scoped Row-Level Security (RLS) guaranteeing 0% data loss across 5+ warehouse nodes.
- **High-Throughput Cloud Microservices:** Built containerized Python and Node.js microservices deployed on AWS (ECS & Lambda) with idempotent webhook ingestion handling sub-500ms execution latency and 99.9% deduplication reliability.
- **Automated Decisioning Engines:** Engineered high-reliability automated inventory reorder and financial allocation calculation backends, cutting reconciliation turnaround from 45 minutes to <30 seconds with auditable transaction logs.
- **High-Availability Infrastructure:** Implemented resilient failover routines, database connection pooling (Supabase / PgBouncer), automated backups, and structured logging across production client instances.
- **Engineering Leadership & Schema Modeling:** Modeled complex enterprise entity-relationship schemas (10k+ active SKUs) while maintaining strict referential integrity, foreign key constraints, and index optimizations.

Backend Developer (Contract) | Marvels Creative Technology March 2025 – June 2025

- **API & Microservice Optimization:** Architected and optimized 20+ RESTful API endpoints and server routes in TypeScript and Node.js, slashing server response latency by ~35% for high-traffic enterprise applications.
- **PostgreSQL Schemas & Query Tuning:** Designed relational database schemas using Prisma ORM; optimized indexes, query execution plans, and join structures to achieve consistent sub-100ms latency.
- **CI/CD & Automated Testing:** Authored comprehensive automated API integration test suites using Jest and Postman in GitHub Actions CI/CD pipelines, achieving 99.5%+ deployment stability.
- **Security & Data Contracts:** Implemented strict request validation schemas (Zod), rate limiting, and JWT authentication middleware to safeguard client endpoints against abuse.

KEY PROJECTS & SYSTEMS

Automated Data Ingestion & Retrieval Pipeline

- **Impact & Scale:** Built and deployed a production data processing microservice indexing 5,000+ records with semantic vector embeddings and structured extraction, cutting per-record processing cost by ~65%.
- **System Reliability:** Engineered signature-verified webhook receivers, automated validation routines, and idempotent PostgreSQL persistence with transactional rollbacks.

Tech: Python 3.12, Node.js, PostgreSQL, Vector Embeddings, REST APIs, Webhooks, AWS

Distributed Agent Command Center & Tunneling Gateway

- **Impact & Architecture:** Engineered a resilient headless worker daemon bridging messaging queues with remote execution nodes; integrated automated Playwright visual verification checkpoints, health checks, and dynamic Cloudflare HTTPS tunneling.

Tech: Python, Docker, Cloudflare Tunneling, Playwright, Linux, REST APIs

TECHNICAL SKILLS

Languages & Core:	Python (3.11/3.12), TypeScript, JavaScript (Node.js), SQL, Bash / Shell
Backend & APIs:	Node.js, Express, Fastify, Python (FastAPI), RESTful APIs, Webhooks, Idempotent Consumers, Microservices
Databases & Caching:	PostgreSQL (RLS, Migrations, Index Tuning, Partitioning), Supabase, Prisma ORM, Redis, PgBouncer
Cloud & DevOps:	AWS (ECS, Lambda, S3, CloudWatch), Docker, GitHub Actions (CI/CD), Render, Cloudflare, Linux Admin
Testing & Tooling:	Jest, PyTest, Postman, Zod, Git, Cursor, Claude